THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $\frac{39}{65}$	Reduce the fraction; only 2 and 5 may remain in the denominator.
2	**(c)** 4	$400 = 2^4 \times 5^2$
3	**(c)** $\frac{4}{33}$	$x = \frac{ab}{99}$, then reduce.
4	**(a)** 6	Find how many digits repeat.
5	**(c)** 6	Multiply under the root.
6	**(a)** rational	Collect the rational and irrational parts.
7	**(b)** $\frac{3}{7}$	Take the average.
8	**(d)** 11	Apply Brahmagupta's rules in order.
9	**(c)** $\frac{21}{70}$	Reduce the fraction; only 2 and 5 may remain in the denominator.
10	**(c)** 3	$250 = 2^1 \times 5^3$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Multiply by 10 and by 100 and subtract: $90x = 75$ (1); $x = \frac{5}{6}$ (1).	1+1
12	Long division of 5 by 22 (1); $= 0.2\overline{27}$, block 27 (1).	1+1
13	$= 8^2 - (\sqrt{5})^2$ (1); $= 59$ (1).	1+1
14	Multiply by 10 and by 100 and subtract: $90x = 55$ (1); $x = \frac{11}{18}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$1000x - x = 27$ (1); $999x = 27$ (1); $x = \frac{1}{37}$ (1).	1+1+1
16	$x = 1 + 0.2\overline{3}$ (1); $0.2\overline{3} = \frac{7}{30}$ (1); $x = \frac{37}{30}$ (1).	1+1+1
17	Lowest terms: $\frac{7}{25}$ (1); Denominator $= 25 = 5^2$ (1); Terminates, since only 2 and 5 occur (1).	1+1+1
	OR: $1000x - x = 135$ (1); $999x = 135$ (1); $x = \frac{5}{37}$ (1).	1+1+1
18	$x = 2 + 0.4\overline{6}$ (1); $0.4\overline{6} = \frac{7}{15}$ (1); $x = \frac{37}{15}$ (1).	1+1+1
	OR: Lowest terms: $\frac{13}{80}$ (1); Denominator $= 80 = 2^4 \times 5$ (1); Terminates, since only 2 and 5 occur (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let $x = 1.1\overline{6}$; $10x$ and $100x$ give $90x = 105$ (1); $x = \frac{7}{6}$ (1); Long division: $1.1\overline{6}$ (1); Matches the given decimal (2).	1+1+1+2
	OR: $(3+\sqrt{5})^2 = 14 + 6\sqrt{5}$ (1); $(3-\sqrt{5})^2 = 14 - 6\sqrt{5}$ (1); Sum $= 28$ (1); The surd terms cancel (1); Result is rational (1).	1+1+1+1+1
20	Let $x = 2.3\overline{3}$; $10x$ and $100x$ give $90x = 210$ (1); $x = \frac{7}{3}$ (1); Long division: $2.\overline{3}$ (1); Matches the given decimal (2).	1+1+1+2
	OR: $(4+\sqrt{3})^2 = 19 + 8\sqrt{3}$ (1); $(4-\sqrt{3})^2 = 19 - 8\sqrt{3}$ (1); Sum $= 38$ (1); The surd terms cancel (1); Result is rational (1).	1+1+1+1+1

Mistakes to avoid:

- Testing the denominator of $\frac{p}{q}$ for factors 2 and 5 before reducing the fraction to lowest terms.
- Believing the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{8} = 4$.
- Treating $\frac{22}{7}$ or 3.14 as the exact value of π; both are rational approximations, π itself is irrational.
- Saying a decimal is irrational because it is long or non-terminating; a repeating decimal such as $0.\overline{142857}$ is rational.
- Using $0.\overline{ab} = \frac{ab}{100}$ instead of $\frac{ab}{99}$ when converting a recurring decimal.
- Getting the sign wrong for debt × debt: the product of two negatives is a fortune (positive).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $\frac{6}{11}$	$x = \frac{ab}{99}$, then reduce.
2	**(a)** 2	Find how many digits repeat.
3	**(c)** 10	Multiply under the root.
4	**(b)** rational	Collect the rational and irrational parts.
5	**(c)** $\frac{5}{11}$	Take the average.
6	**(b)** 26	Apply Brahmagupta's rules in order.
7	**(d)** $\frac{51}{85}$	Reduce the fraction; only 2 and 5 may remain in the denominator.
8	**(c)** 5	$4000 = 2^5 \times 5^3$
9	**(b)** $\frac{28}{33}$	$x = \frac{ab}{99}$, then reduce.
10	**(c)** 6	Find how many digits repeat.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Long division of 7 by 36 (1); $= 0.19\overline{4}$, block 4 (1).	1+1
12	$= 9^2 - (\sqrt{7})^2$ (1); $= 74$ (1).	1+1
13	Multiply by 10 and by 100 and subtract: $90x = 83$ (1); $x = \frac{83}{90}$ (1).	1+1
14	Long division of 17 by 30 (1); $= 0.5\overline{6}$, block 6 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$1000x - x = 148$ (1); $999x = 148$ (1); $x = \frac{4}{27}$ (1).	1+1+1
16	$x = 3 + 0.1\overline{6}$ (1); $0.1\overline{6} = \frac{1}{6}$ (1); $x = \frac{19}{6}$ (1).	1+1+1
17	Lowest terms: $\frac{2}{9}$ (1); Denominator $= 9 = 3^2$ (1); Does not terminate: a prime factor other than 2 and 5 occurs (1).	1+1+1
	OR: $1000x - x = 259$ (1); $999x = 259$ (1); $x = \frac{7}{27}$ (1).	1+1+1
18	$x = 1 + 0.5\overline{8}$ (1); $0.5\overline{8} = \frac{53}{90}$ (1); $x = \frac{143}{90}$ (1).	1+1+1
	OR: Lowest terms: $\frac{3}{10}$ (1); Denominator $= 10 = 2 \times 5$ (1); Terminates, since only 2 and 5 occur (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let $x = 3.2\overline{5}$; $10x$ and $100x$ give $90x = 293$ (1); $x = \frac{293}{90}$ (1); Long division: $3.2\overline{5}$ (1); Matches the given decimal (2).	1+1+1+2
	OR: $(2 + \sqrt{7})^2 = 11 + 4\sqrt{7}$ (1); $(2 - \sqrt{7})^2 = 11 - 4\sqrt{7}$ (1); Sum $= 22$ (1); The surd terms cancel (1); Result is rational (1).	1+1+1+1+1
20	Let $x = 1.4\overline{7}$; $10x$ and $100x$ give $90x = 133$ (1); $x = \frac{133}{90}$ (1); Long division: $1.4\overline{7}$ (1); Matches the given decimal (2).	1+1+1+2
	OR: $(5 + \sqrt{2})^2 = 27 + 10\sqrt{2}$ (1); $(5 - \sqrt{2})^2 = 27 - 10\sqrt{2}$ (1); Sum $= 54$ (1); The surd terms cancel (1); Result is rational (1).	1+1+1+1+1

Mistakes to avoid:

- Testing the denominator of $\frac{p}{q}$ for factors 2 and 5 before reducing the fraction to lowest terms.
- Believing the product (or sum) of two irrational numbers is always irrational: $\sqrt{2} \times \sqrt{8} = 4$.
- Treating $\frac{22}{7}$ or 3.14 as the exact value of π; both are rational approximations, π itself is irrational.
- Saying a decimal is irrational because it is long or non-terminating; a repeating decimal such as $0.\overline{142857}$ is rational.
- Using $0.\overline{ab} = \frac{ab}{100}$ instead of $\frac{ab}{99}$ when converting a recurring decimal.
- Getting the sign wrong for debt × debt: the product of two negatives is a fortune (positive).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **World of Numbers** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 6	Multiply under the root.
2	**(d)** irrational	Collect the rational and irrational parts.
3	**(c)** $\frac{1}{3}$	Take the average.
4	**(a)** -3	Apply Brahmagupta's rules in order.
5	**(a)** $\frac{63}{140}$	Reduce the fraction; only 2 and 5 may remain in the denominator.
6	**(d)** 6	$62500 = 2^2 \times 5^6$
7	**(a)** $\frac{35}{99}$	$x = \frac{ab}{99}$, then reduce.
8	**(b)** 3	Find how many digits repeat.
9	**(c)** 14	Multiply under the root.
10	**(c)** irrational	Collect the rational and irrational parts.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$= 6^2 - (\sqrt{11})^2$ (1); $= 25$ (1).	1+1
12	Multiply by 10 and by 100 and subtract: $90x = 17$ (1); $x = \frac{17}{90}$ (1).	1+1
13	Long division of 11 by 18 (1); $= 0.6\overline{1}$, block 1 (1).	1+1
14	$= 10^2 - (\sqrt{3})^2$ (1); $= 97$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$1000x - x = 223$ (1); $999x = 223$ (1); $x = \frac{223}{999}$ (1).	1+1+1
16	$x = 4 + 0.3\overline{3}$ (1); $0.3\overline{3} = \frac{1}{3}$ (1); $x = \frac{13}{3}$ (1).	1+1+1
17	Lowest terms: $\frac{7}{60}$ (1); Denominator $= 60 = 2^2 \times 3 \times 5$ (1); Does not terminate: a prime factor other than 2 and 5 occurs (1).	1+1+1
	OR: $1000x - x = 481$ (1); $999x = 481$ (1); $x = \frac{13}{27}$ (1).	1+1+1
18	$x = 2 + 0.7\overline{5}$ (1); $0.7\overline{5} = \frac{34}{45}$ (1); $x = \frac{124}{45}$ (1).	1+1+1
	OR: Lowest terms: $\frac{19}{400}$ (1); Denominator $= 400 = 2^4 \times 5^2$ (1); Terminates, since only 2 and 5 occur (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Let $x = 2.6\overline{1}$; $10x$ and $100x$ give $90x = 235$ (1); $x = \frac{47}{18}$ (1); Long division: $2.6\overline{1}$ (1); Matches the given decimal (2).	1+1+1+2
	OR: $(6 + \sqrt{7})^2 = 43 + 12\sqrt{7}$ (1); $(6 - \sqrt{7})^2 = 43 - 12\sqrt{7}$ (1); Sum $= 86$ (1); The surd terms cancel (1); Result is rational (1).	1+1+1+1+1
20	Let $x = 4.5\overline{2}$; $10x$ and $100x$ give $90x = 407$ (1); $x = \frac{407}{90}$ (1); Long division: $4.5\overline{2}$ (1); Matches the given decimal (2).	1+1+1+2
	OR: $(1 + \sqrt{11})^2 = 12 + 2\sqrt{11}$ (1); $(1 - \sqrt{11})^2 = 12 - 2\sqrt{11}$ (1); Sum $= 24$ (1); The surd terms cancel (1); Result is rational (1).	1+1+1+1+1

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